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Criterion Validity of the Psychopathy Checklist in Legal Contexts: An Updated Meta-Analysis

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ABSTRACT

The Psychopathy Checklist (PCL) scales are the most widely used instruments in the assessment of psychopathic personality traits. This updated meta-analysis represents the most extensive synthesis of the PCL scales' criterion validity regarding recidivism and institutional outcomes in legal contexts. According to the results, the PCL total score and Factor 2 criterion validity were moderate and significantly higher for Factor 2 than the total score. The criterion validity of Factor 1 was small and significantly lower than Factor 2 and the total score. The criterion validity of the PCL total score varied considerably between outcomes, with moderate effect sizes for general and violent recidivism, institutional misconduct and violence, and small effect sizes for sexual and intimate partner violence recidivism. A moderator analysis indicated country and sex as the only significant predictors moderating the PCL scales' criterion validity. The total score and Factor 2 were less predictive in the United States than in Canada and Europe for institutional violence. Factor 1 was more predictive in females than males for general and violent recidivism. The present findings may support decision-makers in interpreting the prognostic value and limitations of the PCL scales in the criminal justice system.

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Psychopathy is a personality condition characterized by a superficial, charming interactional style, remorselessness, lack of empathy, and manipulative, impulsive, irresponsible, and antisocial conduct (Hare & Neumann, 2008). Psychopathy is not a clinical diagnosis in current psychiatric classification systems. It can be regarded as a subtype of antisocial personality disorder according to the DSM-5 (APA, 2013), dissocial personality disorder according to the ICD-10 (WHO, 2020), or the trait domain of dissociality according to the ICD-11 (WHO, 2022). Within correctional and forensic-psychiatric samples, the prevalence of psychopathy has been reported to be higher (28% to 34%; Fox & DeLisi, 2019) compared to the general population (about 1.2%; Sanz-Garcia et al., 2021). Meta-analytical data further reported the prevalence to be two to three times higher in males compared to females (Beryl et al., 2014; Fox & DeLisi, 2019) and higher in North America compared to Europe, though the evidence may be biased by methodological differences (Beryl et al., 2014). Individuals meeting the criteria for psychopathy are thought to be responsible for over half of all severe crimes (Falk et al., 2014; Farrington & Bergström, 2018; Harris et al., 2001; Wolfgang et al., 1972). The risk of committing offenses, particularly violent crimes, has been suggested to be elevated in a small sub-group of highly psychopathic individuals

compared to the general population (Fox et al., 2015). Psychopathy has, therefore, also been described as a concept for a theory of crime associated with a variety of adverse legal and criminal justice outcomes (Fox & DeLisi, 2019; Hare, 2016).

The Psychopathy Checklist (PCL) is one of the most widely used instruments in the assessment of psychopathic personality traits in clinical, research, or legal settings. The original PCL was developed in the 1980s (Hare & Frazelle, 1980) and revised to the PCL-R in the 1990s (Hare, 1991, 2003) by Canadian psychologist Robert D. Hare. The PCL-R is a rating scale containing 20 items coded based on a review of an individual's file information and a semi-structured interview. The interview part is not mandatory and is occasionally omitted, particularly in research settings. Initially, and based on explorative factor analyses, the items were assumed to load onto two correlated factors, Factor 1 (F1) and Factor 2 (F2), respectively. F1 describes the core psychopathic personality traits, including superficiality, habitual lying, manipulation, callousness, as well as lack of affect, empathy, remorse, or guilt, whereas F2 reflects the social deviance component in terms of a chronically irresponsible, impulsive, and criminally versatile lifestyle. The two factors are highly correlated (Harpur et al., 1989), suggesting that

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they result from a common underlying personality condition (Fowles, 2011; Neumann et al., 2015). The PCL-R total score of a maximum of 40 points, which summarizes the two factors, including two additional items that do not load on either factor, can be used to identify the degree of an individual's psychopathic personality traits. Additional scales were derived from the PCL-R, including the shorter Psychopathy Checklist: Screening Version (PCL:SV; Hart et al., 1995) and the Psychopathy Checklist: Youth Version (PCL:YV; Forth et al., 2003). A list of the 20 PCL-R items is provided in the [online supplementary materials](#) (Section S1).

Concerning the PCL-R factors, various confirmatory factor analytic models have been discussed, including a hierarchical three-factor model (e.g., Hare, 2003; Hare & Neumann, 2008) and a correlated four-factor model (e.g., Hare & Neumann, 2005). Alternatively, in a hierarchical manner, F1 and F2 can be deemed as second-order factors (comprising two first-order factors, Interpersonal and Affective or Lifestyle and Antisocial, respectively). Given the frequency with which F1 and F2 have been utilized within PCL-based research on psychopathy, they are also used as a frame of reference in the current study.

Concerning the PCL-R total score, psychometric research suggested cutoffs to identify people with psychopathic characteristics. For males, a North American/Canadian standard PCL-R cutoff score of ≥ 30 (Hare, 1991) and a European standard PCL-R cutoff score of ≥ 25 (Cooke, Hart, et al., 2004; Cooke, Michie, et al., 2004) have been suggested. For females, a PCL-R cutoff score of ≥ 23 has been suggested (Vogel et al., 2014). The downside of such dichotomization is that it reduces the information of a continuum into two categories. Taxometric research, therefore, recommended PCL-R point ranges to gauge psychopathic trait levels, i.e., very low (0–8), low (9–16), moderate (17–24), high (25–32), and very high (33–40; Hare, 2003, p. 31), suggesting that PCL-R scores reflect an underlying dimensional construct rather than a taxon (e.g., Edens et al., 2006; Guay et al., 2007; Walters et al., 2007; Walters, 2014).

Research has shown that the PCL scales predict recidivism and institutional outcomes in forensic samples (Hare et al., 2000; Quinsey et al., 1995; Rice & Harris, 1995; Salekin et al., 1996). They are, therefore, commonly utilized as predictive indicators of risk in the criminal justice system, such as to make sentencing or release decisions (DeMatteo et al., 2014, 2016; DeMatteo et al., 2019). Over the past decades, several meta-analyses have examined the criterion-related validity of the PCL scales. By criterion validity, we refer to the fact that studies typically apply either prospective or retrospective designs. The most recent meta-meta-analysis by DeMatteo and Olver (2022) synthesized evidence on the criterion validity of the instrument across 18 previous meta-analyses, including 208 primary studies published between 1984 and 2019, a reference list of which is provided in the [online supplementary materials](#) (Section S2). Most studies focused on the original PCL, the PCL-R, and the PCL:SV, including a few studies using the PCL:YV. Six outcomes were evaluated, i.e., general recidivism (k meta-analyses, $k_{ma} = 7$), violent recidivism

($k_{ma} = 9$), sexual recidivism ($k_{ma} = 3$), and intimate partner violence (IPV) recidivism ($k_{ma} = 2$), as well as institutional misconduct ($k_{ma} = 4$) and institutional violence ($k_{ma} = 5$). The results of the meta-meta-analysis suggested the PCL scales' total score and F2 to have moderate criterion validity for most outcome criteria in terms of Cohen's d (Cohen, 1988), including general recidivism, violent recidivism, IPV recidivism, and institutional misconduct (Cohen's $d = 0.5$ – 0.8), small to moderate criterion validity for institutional violence (Cohen's $d = 0.2$ – 0.8), and small criterion validity for sexual recidivism (Cohen's $d = 0.2$ – 0.5). F1 was suggested to have a small predictive value concerning all outcomes (Cohen's $d = 0.2$ – 0.5), except for a negligible effect for sexual recidivism (Cohen's $d < 0.2$).

The first aim of the present work was to update the evidence on the criterion validity of the PCL scales. This update was based on all original primary studies covered in the meta-meta-analysis by DeMatteo and Olver (2022). In addition, we aimed to conduct a database search to collect eligible studies that might have been published before the latest release date of the original primary studies in 2019, as well as additional studies that might have been published after that release date.

The second aim was to extend the analysis by DeMatteo and Olver (2022) by conducting a moderator analysis on relevant predictors that may explain heterogeneity in the criterion validity of the PCL scales. Although DeMatteo and Olver (2022) did not report a moderator analysis themselves, five of the 18 meta-analyses did so (Guy et al., 2005; Hawes et al., 2013; Leistico et al., 2008; Walters, 2003b; Yang et al., 2010). Based on these considerations, a moderator analysis using an information-theoretic approach was planned along with the updated meta-analysis considering 12 study-, sample-, and individual-specific predictors. Taken together, the present meta-analysis aimed to provide an update on the evidence on the criterion validity of the PCL scales and the predictors moderating its strength.

Methods

Study selection

The first author, L.H., a senior researcher, conducted the study selection procedure. L.H. collected all original publications, performed the database search, and evaluated potential studies according to the inclusion criteria. L.H. and two additional raters, C.E. and F.F., both medical students, then independently double-coded the data from the publications. L.H. coded all studies. C.E. and F.F. each coded half of the studies. L.H. then revisited cases of disagreement in the extracted data and made a final decision.

The inclusion criteria largely followed those proposed by DeMatteo and Olver (2022). Eligible studies had (1) to be conducted in adults, (2) to use the PCL-R or the PCL:SV, (3) to report a measure of criterion validity concerning recidivism outcomes defined by official recidivism records, or (4) to report a measure of criterion validity concerning institutional outcomes defined as either institutional misconduct (i.e., verbal aggression or other

destructive behavior) or institutional violence (i.e., physical aggression).

First, the 208 primary studies covered in the meta-meta-analysis by DeMatteo and Olver (2022) were checked for accessibility. Nine primary publications were inaccessible conference contributions, unpublished manuscripts, or institutional reports, which were unavailable online, *via* an academic library, or by request to the authors and were, thus, not included. Another 32 studies described in dissertations, conference contributions, institutional reports, or unpublished manuscripts were only accessible in abstract form, but the predictive measures could be extracted from the previous meta-analyses. A further 26 studies conducted in juvenile samples or using the PCL:YV were excluded because the present work considered adult samples only. Two studies reporting PCL-R facets only or not providing information on criterion validity were excluded. The data of ten unique samples reported in more than one publication was included based on the evidence reported in the most recent publication; for two samples, the current authors identified more recent publications. Together, this *original* dataset consisted of 181 unique samples identified from 171 records comprising 35,272 individuals published between 1984 and 2019.

Second, a database search was conducted to identify additional eligible studies. The databases searched included the Cochrane Library, MedLine, Google Search, and Dissertation Abstracts International. The search used the term “psychop*” together with the Boolean terms “recidivi*” OR “reoffend*” OR “infraction*” OR “misconduct*.” This search identified an *additional* dataset of 46 unique samples from 46 records comprising 11,585 individuals published between 2000 and 2022. Thirty-three of these studies were published before the latest release date of the original primary studies in 2019, but there was also a considerable number of 13 studies published after that release date.

Finally, the *original* and *additional* datasets were compiled into the *updated* dataset. The updated dataset consisted of 217 samples from 227 records comprising 46,857 individuals published between 1984 and 2022. A flowchart detailing the study selection as well as a reference list of all studies included in the updated dataset are provided in the [online supplementary materials](#) (Section S2).

Meta-analysis

The data was analyzed using R Statistical Software (version 2024.04.2; R Core Team, 2022). The updated meta-analysis was conducted using the metafor package (Viechtbauer, 2010). A multilevel random-effects model was applied to account for the nested data structure, where outcome- and factor-specific effect sizes were nested within samples. A five-level model was chosen (i.e., the outcome, factor, cluster, study, and subject level). The cluster level accounted for the assumption that some of the observed effect size estimates could not be independent because they were computed based on at least partially overlapping samples. To implement the cluster level, the variance-covariance matrix of the

studies with likely overlapping samples was approximated using the vcalc function in the metafor package (Viechtbauer, 2010). Since the variance-covariance matrix is often just a rough approximation and since the random effects structure may not fully capture all dependencies in the underlying true effects, the cluster-robust inference method was added to the model using the clubSandwich package (Bell & McCaffrey, 2002; Pustejovsky & Tipton, 2018). Residual heterogeneity was reported in terms of Q and I^2 .

Six outcomes were evaluated, namely general recidivism ($k = 98$ samples; 45% of the total samples), violent recidivism ($k = 114$; 53%), sexual recidivism ($k = 46$; 21%), institutional misconduct ($k = 32$; 15%), institutional violence ($k = 54$; 25%), and IPV recidivism ($k = 7$; 3%). Approximately half of the studies (51%) reported more than one outcome. Each outcome was evaluated concerning the PCL scales' total score ($k = 215$; 99%) and F1/F2 ($k = 106$; 49%), if reported in the primary studies. If more than one predictive measure was reported for an outcome-factor combination in the primary studies, the aggregated effect size was reported in the present work.

Effect sizes were meta-analyzed using point-biserial correlation coefficients (r_{pb}). If primary studies reported other effect sizes in terms of Cohen's d , the area under the curve (AUC), or the diagnostic odds ratio (DOR), a base-rate-corrected conversion (Salgado, 2018) was applied to account for the potential impact of base rates on the magnitude of r_{pb} (Rice & Harris, 2005). The correction considered the mean base rate per outcome and gender because a priori analysis suggested evident variation in base rates depending on outcome and gender. Females compared to males had significantly lower base rates of general recidivism (females 30% vs. males 47%, $t = -3.20$, $p = .002$), violent recidivism (females 20% vs. males 30%, $t = -2.63$, $p = .009$), and institutional misconduct (females 16% vs. males 39%, $t = -2.49$, $p = .017$), but higher base rates of institutional violence (females 53% vs. males 37%, $t = 2.57$, $p = .012$). There were no sex-related differences in base rates for sexual recidivism (mean 18%) or IPV recidivism (mean 27%), as only males were assessed regarding these outcomes. Samples with missing base rates could not be corrected.

Pairwise comparisons between the outcome-specific effect sizes for the PCL scales' total score, F1 and F2, were computed based on general linear hypothesis (GLH) testing using the glht command in the multcomp package. The Bonferroni correction was applied to control for the family-wise error rate (FWER). The Bonferroni corrected p -value, computed as $p_{corrected} = 1 - (1 - p_{original} / k)^k$, where $p_{original} = .05$ indicates the original p -value, and k the number of dependent variables ($n = 3$), was $p_{corrected} = .049$. Because some researchers suggested the Bonferroni correction to be too conservative in models with latent factors (Cribbie, 2000; Ozenne et al., 2023), a sensitivity analysis using the false discovery rate (FDR) was conducted as a possible compromise to control for the FWER (Cribbie, 2007). The sensitivity analysis, however, revealed only marginally different findings, which are only provided in the [online supplementary materials](#) (Section S3).

Moderator characteristic

The characteristics of the moderators were extracted from the primary studies. All moderators were represented as categorical variables. Continuous variables were categorized to avoid multicollinearity potentially undermining the statistical significance of multimodel approaches (Allan, 1997). Missing moderator values were classified as *unclear*.

The following 12 study-, sample-, and individual-specific moderators, with the subgroups labels in brackets, were assessed: (a) publication status [published, unpublished], (b) publication year [<2000, 2000–2010, >2010], (c) country of origin [Canada, Europe, United Kingdom, United States, other], (d) design of the study [prospective, retrospective, unclear], (e) source of information [file, file/interview, unclear], (f) version of the PCL scale [PCL-R, PCL: SV], (g) inter-rater reliability of the PCL scales' total score [poor, moderate, good, excellent, unclear], (h) sample size [<100, 100–400, >400], (i) setting of data collection [prison, hospital, unclear], (j) follow-up length [<1 year, 1–5 years, >5 years, unclear], (k) age [≤35 years, >35 years], and (l) sex [females, males, mixed, males and mixed]. Details on the moderator coding scheme are provided in the [online supplementary materials](#) (Section S4).

Moderator analysis

Moderator analyses involving multiple predictors typically require numerous regression models. The main drawback of standard multiple regression models is that the statistical significance of a given moderator may vary substantially between models, depending on which other moderators are included. Standard multiple regression, therefore, requires methodological decisions for selecting appropriate moderator combinations step by step (Whittingham et al., 2006). This is cumbersome or unfeasible if the number of predictors is high, as in the present analysis. To overcome this limitation, an information-theoretic approach applying multimodel inference to generalize several or even all possible statistical models to estimate the relative importance of multiple moderators has been suggested as a suitable alternative (Burnham & Anderson, 2002).

The multimodel moderator analysis was carried out using the *metafor* package (Viechtbauer, 2010) in combination with the *glmulti* package (Calcagno & Mazancourt, 2010). As for the main meta-analysis, the basic model accounted for the nested structure of the data using a multilevel random-effects model. An exhaustive model screening was conducted, considering all possible combinations between the 12 moderators, resulting in a set of 4,096 models. Model fit was estimated by weighting each model based on the Akaike Information Criterion corrected for small sample size (AICc) (Calcagno & Mazancourt, 2010). The AICc was preferred over the traditional AIC because the latter may overfit small sample sizes, potentially producing bias in multi-term models (Claeskens & Hjort, 2008). For the AICc values being directly comparable between models, heterogeneity was estimated based on the maximum-likelihood estimator

(ML) (Gurka, 2006). The resulting AICc weight for a given model thus represented the probability of being the optimal model out of all models (Viechtbauer, 2022). Based on this weighting procedure, the best 300 models, referred to as the candidate models, were considered for further analysis. Relative moderator importance was estimated based on the summed AICc weights across the candidate models in which a given moderator appeared. Moderators appearing in larger-weight models typically receive higher importance values (IM). Moderator's IM reflects its overall support across candidate models, ranging from 0 to 1. A cutoff of IM = .8 was chosen, sometimes used to differentiate between important and less important predictors, though this is an arbitrary division and serves illustrative purposes (Calcagno & Mazancourt, 2010). Last, model-averaged moderator effect sizes regarding Cohen's *d* and its corresponding confidence intervals (95% CI) were computed using multimodel inference across the candidate models. These effect sizes are called *unconditional estimates* as they are not contingent upon only one model but are generalized across the candidate models. The significance of contrasts between the effect sizes moderator subgroups was assessed based on general linear hypothesis (GLH) testing using the *multcomp* package (Hothorn et al., 2008), with the Bonferroni correction to counteract the problem of multiple comparisons. Using the abovementioned formula, the Bonferroni corrected *p*-value also revealed a rounded $p_{corrected} = .049$ in all instances, accounting for different moderator subgroups ($k=2, 3, 4$, or 5).

Publication bias

To assess potential publication bias, Egger's test (Egger et al., 1997) and Begg and Mazumdar's test for rank correlation (Begg & Mazumdar, 1994) assessing funnel plot asymmetry, as well as a *p*-curve analysis (Simonsohn et al., 2014) were computed.

Results

Meta-analysis

Results of the updated meta-analysis are listed in [Table 1](#) and illustrated in the forest plot in [Figure 1](#). The outcome- and factor-specific effect sizes regarding the criterion validity of the PCL scales, which are reported in terms of Cohen's *d* with the 95% CI, can be interpreted as the degree to which they accurately predict a given outcome. The strength of the effect sizes is indicated following Cohen's guidelines, with $d \geq 0.2$ indicating small, $d \geq 0.5$ moderate, and $d \geq 0.8$ large criterion validity (Cohen, 1988). For example, Cohen's $d = 0$ would indicate that the PCL instruments predict no difference in the standard deviation (SD) between the means of recidivists and non-recidivists, whereas $d = 1$ would suggest that the PCL instruments predict a difference between the means of recidivists and non-recidivists of one SD. For a tool to be useful within the context of risk assessment, a criterion validity with an effect size of least $d \geq 0.358$

Table 1. Meta-analysis of the predictive validity of the PCL scales.

Outcome/Factor level	Main analysis				Sensitivity analysis									
	<i>k</i>	<i>n</i>	<i>d</i> [95% CI]	<i>p</i>	Prospective				Retrospective				Pairwise	
					<i>k</i>	<i>n</i>	<i>d</i> [95% CI]	<i>p</i>	<i>k</i>	<i>n</i>	<i>d</i> [95% CI]	<i>p</i>	<i>z</i>	<i>p</i>
All outcomes F1	106	22,601	0.30 [0.25, 0.34]	<.001	40	8,388	0.31 [0.23, 0.39]	<.001	58	13,392	0.29 [0.23, 0.35]	<.001	-0.43	1.000
All outcomes F2	106	22,601	0.60 [0.55, 0.65]	<.001	40	8,388	0.58 [0.50, 0.67]	<.001	58	13,392	0.61 [0.54, 0.68]	<.001	0.46	1.000
All outcomes Total	215	45,199	0.55 [0.50, 0.59]	<.001	80	15,468	0.52 [0.44, 0.60]	<.001	114	26,757	0.56 [0.50, 0.62]	<.001	0.77	1.000
General recidivism F1	48	12,187	0.34 [0.28, 0.39]	<.001	18	4,457	0.29 [0.20, 0.39]	<.001	28	7,425	0.36 [0.29, 0.44]	<.001	1.12	1.000
General recidivism F2	48	12,187	0.77 [0.71, 0.84]	<.001	18	4,457	0.66 [0.55, 0.76]	<.001	28	7,425	0.84 [0.75, 0.93]	<.001	2.59	1.000
General recidivism Total	96	20,853	0.64 [0.59, 0.70]	<.001	33	6,711	0.58 [0.49, 0.67]	<.001	51	11,872	0.68 [0.61, 0.76]	<.001	1.66	1.000
Violent recidivism F1	55	13,066	0.33 [0.28, 0.39]	<.001	19	4,833	0.29 [0.20, 0.39]	<.001	33	7,827	0.34 [0.27, 0.42]	<.001	0.82	1.000
Violent recidivism F2	55	13,066	0.64 [0.58, 0.70]	<.001	19	4,833	0.61 [0.51, 0.72]	<.001	33	7,827	0.66 [0.58, 0.74]	<.001	0.74	1.000
Violent recidivism Total	113	26,927	0.56 [0.51, 0.61]	<.001	35	8,541	0.49 [0.40, 0.58]	<.001	68	17,285	0.59 [0.52, 0.66]	<.001	1.60	1.000
Sexual recidivism F1	23	6,137	0.11 [0.05, 0.17]	<.001	3	734	0.27 [0.08, 0.48]	.006	19	5,330	0.12 [0.05, 0.20]	.001	-1.39	1.000
Sexual recidivism F2	23	6,137	0.28 [0.21, 0.34]	<.001	3	734	0.15 [-0.05, 0.35]	.137	19	5,330	0.31 [0.23, 0.39]	<.001	1.47	1.000
Sexual recidivism Total	46	12,089	0.38 [0.32, 0.43]	<.001	8	1,816	0.31 [0.18, 0.45]	<.001	37	10,200	0.41 [0.33, 0.48]	<.001	1.15	1.000
Institutional misconduct F1	19	2,248	0.44 [0.33, 0.55]	<.001	12	1,490	0.43 [0.30, 0.57]	<.001	6	703	0.43 [0.24, 0.63]	<.001	0.01	1.000
Institutional misconduct F2	19	2,248	0.48 [0.37, 0.59]	<.001	12	1,490	0.50 [0.36, 0.64]	<.001	6	703	0.43 [0.24, 0.63]	<.001	-0.58	1.000
Institutional misconduct Total	30	3,864	0.58 [0.49, 0.67]	<.001	19	2,644	0.58 [0.46, 0.71]	<.001	10	1,165	0.57 [0.42, 0.73]	<.001	-0.13	1.000
Institutional violence F1	33	4,471	0.30 [0.23, 0.38]	<.001	15	2,240	0.33 [0.21, 0.45]	<.001	13	1,788	0.29 [0.17, 0.40]	<.001	-0.48	1.000
Institutional violence F2	33	4,471	0.51 [0.43, 0.59]	<.001	15	2,240	0.56 [0.44, 0.70]	<.001	13	1,788	0.45 [0.33, 0.57]	<.001	-1.27	1.000
Institutional violence Total	53	7,974	0.49 [0.42, 0.56]	<.001	29	4,502	0.50 [0.39, 0.61]	<.001	19	2,901	0.49 [0.38, 0.60]	<.001	-0.14	1.000
Intimate partner violence recidivism F1	2	-	-	.004	-	-	-	-	2	-	-	.003	-	-
Intimate partner violence recidivism F2	2	-	-	.013	-	-	-	-	2	-	-	.009	-	-
Intimate partner violence recidivism Total	7	1,291	0.42 [0.25, 0.61]	<.001	-	-	-	-	7	1,291	0.45 [0.27, 0.64]	<.001	-	-
Total	227	46,857			80	15,781			116	27,852			-	-

Note. The outcome- and factor-specific effect sizes from the updated meta-analysis are listed. Given is Cohen's *d* (*d* [95% confidence intervals, CI]) for Factor 1 (F1), Factor 2 (F2), and the total score concerning the criterion validity of the PCL scales. Information on the number of samples (*k*) and number of individuals (*n*) is provided for each outcome and factor level. Sensitivity analysis: The outcome- and factor-specific effect sizes are also listed separately for prospective and retrospective study designs and the pairwise comparisons concerning the study design. Z-values smaller than zero favor the first-mentioned subgroup, and vice versa; z-values larger than zero favor the second-mentioned subgroup. The Bonferroni correction was applied to account for multiple comparisons.

(corresponding to an AUC $\geq .60$) has been recommended (DeMatteo & Olver, 2022). The following sections report all outcome- and factor-specific effect sizes, but the pairwise comparisons (*z*- and *p*-values) between them are only reported in case of statistical significance and otherwise provided in the [online supplementary materials](#) (Section S3).

Across all outcomes, the criterion validity of the PCL scales' total score was found to be moderate ($d = 0.55$, 95% CI [0.50, 0.59]). F2 also had a moderate predictive value ($d = 0.60$, 95% CI [0.55, 0.65]) that was, however, significantly higher than the total score (pairwise total score vs. F2: $z = -4.00$, $p < .001$). F1, by contrast, had a small predictive value ($d = 0.30$, 95% CI [0.25, 0.34]) and was significantly smaller than both the total score (pairwise total score vs. F1: $z = 18.76$, $p < .001$) and F2 (pairwise F2 vs. F1: $z = 22.61$, $p < .001$).

Within outcomes, the PCL total score had moderate or close-to-moderate criterion validity for most outcome criteria, including general recidivism ($d = 0.64$, 95% CI [0.59, 0.70]), violent recidivism ($d = 0.56$, 95% CI [0.51, 0.61]), institutional misconduct ($d = 0.58$, 95% CI [0.49, 0.67]), and institutional violence ($d = 0.49$, 95% CI [0.42, 0.56]). By contrast, the predictive estimates for sexual recidivism ($d = 0.38$, 95% CI [0.32, 0.43]) and IPV recidivism ($d = 0.42$, 95% CI [0.25, 0.61]) were small. Although the effect size for sexual recidivism was significantly smaller compared to most other outcomes (all pairwise $p_{corrected} < .049$, except compared

to institutional violence), the estimate for IPV recidivism did not differ significantly from any of the other outcomes due to the significant variance in the small number of samples available for this outcome (all pairwise $p_{corrected} \geq .049$).

F2 was comparable in criterion validity to the total score because moderate or close-to-moderate predictive values were identified for most outcomes. More specifically, slightly but significantly larger effect sizes than for the total score were observed for general recidivism ($d = 0.77$, 95% CI [0.71, 0.84]; pairwise total score vs. F2: $z = -5.58$, $p < .001$) and violent recidivism ($d = 0.64$, 95% CI [0.58, 0.70]; pairwise total score vs. F2: $z = -3.82$, $p = .020$), but not for institutional misconduct ($d = 0.48$, 95% CI [0.37, 0.59]) or institutional violence ($d = 0.51$, 95% CI [0.43, 0.59]). Again, the predictive value for sexual recidivism ($d = 0.28$, 95% CI [0.21, 0.34], all pairwise $p_{corrected} < .049$) and IPV recidivism ($d = 0.33$, 95% CI [0.07, 0.61], all pairwise $p_{corrected} \geq .049$) was small.

F1 had a smaller criterion validity than the total score and F2. This observation reached statistical significance when comparing the two factors concerning general, violent, sexual recidivism, and institutional violence (pairwise total score/F2 vs. F1 $p_{corrected} < .049$). In particular, F1 evinced small effects for most outcome criteria, including general recidivism ($d = 0.34$, 95% CI [0.28, 0.39]), violent recidivism ($d = 0.33$, 95% CI [0.28, 0.39]), institutional misconduct ($d = 0.44$, 95% CI [0.33, 0.55]), institutional violence ($d = 0.30$, 95% CI [0.23, 0.38]), and IPV recidivism ($d = 0.38$, 95% CI [0.12, 0.67]).

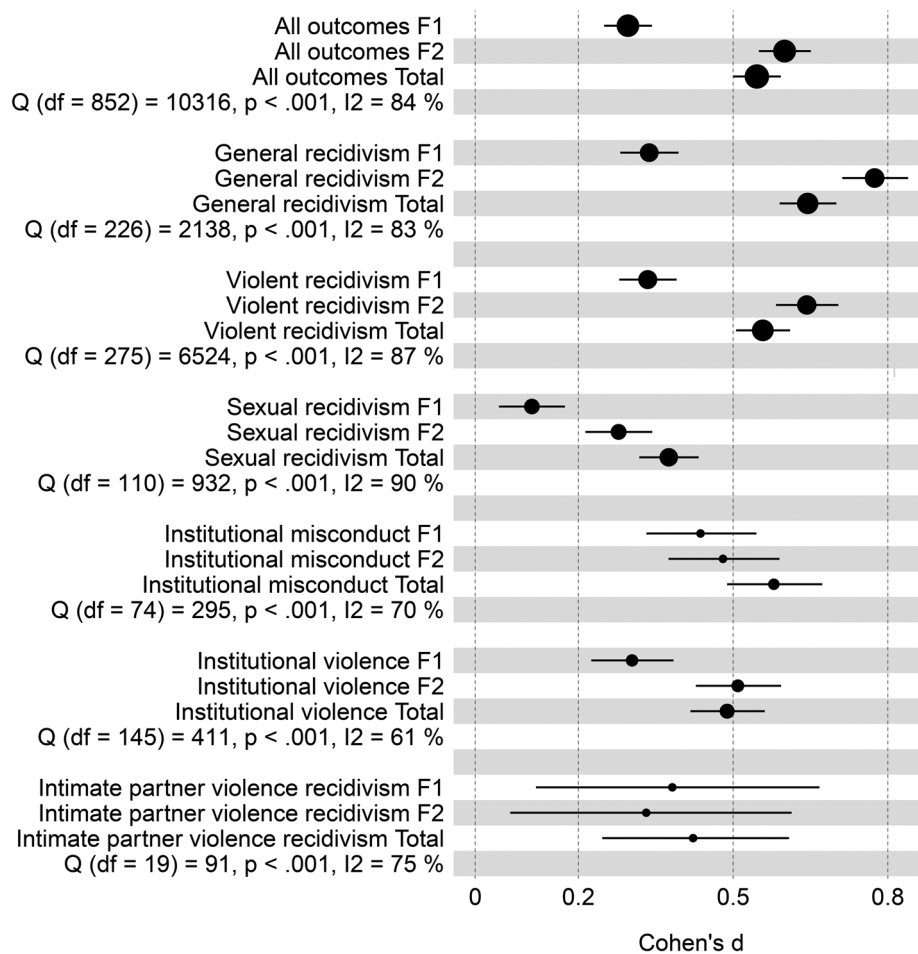


Figure 1. Forest plot. Note. Forest plot illustrating the outcome- and factor-specific effect sizes in terms of Cohen's d with 95% confidence intervals (CI) concerning the criterion validity of PCL scales' Factor 1 (F1), Factor 2 (F2), and the total score obtained in the updated meta-analysis. The circle size is proportionate to the precision of the outcome-factor-specific effect sizes. Vertical dashed lines reference the strength of the effect sizes according to Cohen's guidelines ($d \geq 0.2$ small, $d \geq 0.5$ moderate, $d \geq 0.8$ large) (Cohen, 1988). The residual heterogeneity (Q) and I^2 for each outcome are also given.

In contrast, the effect for sexual recidivism was negligible ($d = 0.11$, 95% CI [0.05, 0.17], all pairwise $p_{corrected} < .049$).

The residual heterogeneity observed in the updated meta-analysis was substantial when estimated across all outcomes, $Q (df = 854) = 11,620$, $p < .001$, $I^2 = 84\%$. Although there was a tendency for the recidivism outcomes to be more heterogeneous compared to the institutional outcomes, none of the outcome-specific effects explained much of the heterogeneity, as depicted in the forest plot (Figure 1). The sizeable residual heterogeneity indicated that a substantial percentage of the observed variance could not be attributed to sampling error but must be considered systematic differences between studies. These observations corroborated the importance of a thorough moderator analysis that could explain variation in the criterion validity of the PCL, as reported in the subsequent sections.

Moderator characteristic

The characteristics of 12 study-, sample-, and individual-specific moderators considered in the updated meta-analysis are listed in Table 2. The table indicates the number and frequency of the moderator subgroups across all outcomes,

as detailed below. Details on the moderator characteristics for the separate outcomes are provided in the [online supplementary materials](#) (Section S4).

The publication status was considered published if studies were reported in scientific journals or books, which was the case for most studies ($k = 173$; 80%). The remaining studies ($k = 44$; 20%) were considered unpublished in terms of institutional reports, conference contributions, theses, dissertations, or unpublished manuscripts.

The publication year ranged from 1984 to 2022. Most studies were published between 2000 and 2010 ($k = 117$; 54%), approximately one fifth was published before 2000 ($k = 53$; 24%) or after 2010 ($k = 47$; 22%).

The country of origin was divided into five regions. Canada ($k = 101$; 47%) and the United States ($k = 47$; 22%) contributed most of the studies, followed by Europe ($k = 45$; 21%, including Austria, Belgium, Finland, Germany, Netherlands, Portugal, Spain, Sweden, Switzerland, and Turkey), a region summarized under the United Kingdom ($k = 19$; 9%, including the United Kingdom, Scotland, Australia, and New Zealand), and other countries ($k = 5$; 2%, including Brazil, Singapore, and South Korea).

The study design was retrospective in most studies ($k = 116$; 53%) and prospective otherwise ($k = 80$; 37%). In some

cases, the study design was unclear ($k = 21$; 10%). Prospective versus retrospective study designs were distinguished based on the time point of the data collection. Prospective study designs were defined as collecting data from a given time point into the future. Retrospective study designs were defined as collecting data from a given time point in the past. In the case of retrospective designs, studies reported coding PCL data either from file records or using already generated PCL ratings (pseudo-prospective design). Most retrospective studies reported procedures to prevent criterion contamination ($k = 81$; 37%), though some studies did not provide this information ($k = 33$; 15%). Only a few studies ($k = 2$; 1%) reported concerns regarding criterion contamination. A sensitivity analysis assessing whether prospective versus retrospective study designs affected the above-described meta-analytic evidence on the criterion validity concerning the outcomes or factors suggested no significant differences. Details on the sensitivity analysis are provided in Table 1.

The sources of information for scoring the PCL scales were equally frequently reported to be either file-based only ($k = 93$; 43%) or a combination of file reviews and semi-structured interviews ($k = 94$; 43%). In some cases, the source was unclear ($k = 30$; 14%).

The version used to score psychopathy most commonly reported was the PCL-R ($k = 186$; 86%), whereas the PCL:SV ($k = 32$; 15%) was less frequently reported. After converting the mean PCL:SV total scores to PCL-R equivalents (multiplication with the ratio of the maximal scores of the PCL-R/PCL:SV = 40/24; Hare et al., 2013), the mean PCL total score across all samples was 18.98 (SD 4.55), which was distributed across the five point ranges as follows (Hare, 2003): very low ($k = 2$; 0.9%), low ($k = 38$; 18%), moderate ($k = 106$; 49%), high ($k = 22$; 10%), and very high ($k = 1$; 0.5%). The remaining studies did not report PCL scores ($k = 56$; 26%). The PCL mean total score was not considered as moderators in the present analysis to avoid correlational confounding of the criterion validity measures from the same construct.

The inter-rater reliability of the mean PCL total scores was provided by approximately half of the studies ($k = 113$; 52%) and quantified using the intraclass correlation coefficient (ICC; Shrout & Fleiss, 1979). Most studies reported adequate ($ICC \geq .75$; $k = 60$; 28%), good ($ICC \geq .85$; $k = 19$; 9%), or excellent reliability ($ICC \geq .95$; $k = 28$; 13%), whereas only a few reported poor reliability ($ICC < .75$; $k = 6$; 3%), considering the critical ICC values (Duwe & Rocque, 2017). To avoid small subgroups, only the subgroups reported ($k = 113$; 52%) versus unreported ($k = 104$; 48%) were considered in the present analysis.

The sample size varied considerably between studies, ranging from seven to 1,695 individuals, with a mean of 197 individuals across studies. Most studies reported sample sizes between 100 to 400 individuals ($k = 120$; 55%), about a third reported sample sizes smaller than 100 ($k = 82$; 38%), and the remaining studies were conducted with more than 400 individuals ($k = 26$; 12%).

The settings in which studies were conducted were either prisons ($k = 136$; 63%) or forensic-psychiatric hospitals ($k =$

71; 33%). A small number of studies reported either outpatient ($k = 7$) or mixed settings ($k = 2$) or did not provide any information on the setting ($k = 10$; 5%), which were summarized as *unclear* to avoid small subgroups.

Follow-up length also varied considerably between samples. Some studies reported follow-up periods longer than 5 years ($k = 70$; 32%), between 1 to 5 years ($k = 82$; 38%), or shorter than 1 year ($k = 28$; 13%). The variation in follow-up length was partially explained by the considerably more extended follow-up periods across the recidivism outcomes (mean 6.2 years, range 5.0 months to 23.5 years) compared to the institutional outcomes (1.9 years, range 2 wk–8.8 years). Some studies did not provide information on follow-up length ($k = 51$; 24%).

The mean age of the samples was 33.8 years (median = 33.6 years; range 18.6–50.3 years). Most samples had a mean age younger than 35 years ($k = 102$; 47%), and in the remaining studies reported a mean age above 35 years ($k = 68$; 31%). Some studies did not provide information on age ($k = 48$; 22%). The cutoff at 35 years was data-driven close to the mean/median. An alternative cutoff at 25 years, for which a general decrease in criminal behavior had been suggested according to the well-established age-crime curve (e.g., Baglione et al., 2022; Loeber & Farrington, 2014; Ulmer & Steffensmeier, 2014), would have resulted in a very small subgroup with a mean age younger than 25 years ($k = 10$; 5%) and was therefore not pursued.

The sex of the individuals reported in the samples was predominantly male ($k = 183$; 84%). Only one-tenth of the studies were conducted in female samples ($k = 22$; 10%). A few studies reported mixed samples ($k = 18$; 8%). Due to the high percentage of males in these mixed samples (86% males in the mixed samples), a category combining *males and mixed* samples ($k = 201$; 93%) was assessed separately.

The racial composition of the samples was reported in only 38% of the studies ($k = 83$). The mean percentage of Caucasian individuals across these studies was 67% (range 9.3%–100% Caucasians). Given the large number of studies without race information, it was not considered a moderator in the present analysis.

Moderator analysis

The multimodel moderator analysis considered the 12 study-, sample-, and individual-specific moderators. Due to the extensive number of results generated by the multimodel analysis, only the predictors with statistically significant subgroup contrasts are presented in the main body of the text, as these were suggested to moderate the criterion validity of the PCL scales. The model-averaged moderator effect sizes are reported regarding Cohen's d [95% CI] and their significance level (p -value). The pairwise comparisons between moderator subgroups are reported (z -value, p -value) to highlight the strength of the contrasts. The relative importance values of each moderator subgroup across all outcomes are reported in Table 2 and illustrated in Figure 2. Details on all findings of the moderator analysis are provided in the online supplementary materials (Section S4).

Table 2. Moderator characteristics.

Moderator	F1/F2	Total
Publication	IM = 0.29/0.25	IM = 0.40
Published	<i>k</i> = 84 [37%]	<i>k</i> = 172 [76%]
Unpublished	<i>k</i> = 22 [10%]	<i>k</i> = 43 [19%]
Year	IM = 0.13/0.15	IM = 0.62
<2000	<i>k</i> = 22 [10%]	<i>k</i> = 53 [23%]
2000–2010	<i>k</i> = 59 [26%]	<i>k</i> = 116 [51%]
>2010	<i>k</i> = 25 [11%]	<i>k</i> = 46 [20%]
Country	IM = 0.33/1.00	IM = 0.99
Canada	<i>k</i> = 50 [22%]	<i>k</i> = 98 [43%]
Europe	<i>k</i> = 17 [7%]	<i>k</i> = 44 [19%]
United Kingdom	<i>k</i> = 15 [7%]	<i>k</i> = 24 [11%]
United States	<i>k</i> = 20 [9%]	<i>k</i> = 44 [19%]
Other	<i>k</i> = 4 [2%]	<i>k</i> = 5 [2%]
Design	IM = 0.12/0.10	IM = 0.13
Prospective	<i>k</i> = 40 [18%]	<i>k</i> = 80 [35%]
Retrospective	<i>k</i> = 58 [26%]	<i>k</i> = 114 [50%]
Unclear	<i>k</i> = 8 [4%]	<i>k</i> = 21 [9%]
Source	IM = 0.12/0.23	IM = 0.38
File	<i>k</i> = 49 [22%]	<i>k</i> = 94 [41%]
File/Interview	<i>k</i> = 47 [21%]	<i>k</i> = 92 [41%]
Unclear	<i>k</i> = 10 [4%]	<i>k</i> = 29 [13%]
Version	IM = 0.69/0.65	IM = 0.63
PCL-R	<i>k</i> = 90 [40%]	<i>k</i> = 184 [81%]
PCL:SV	<i>k</i> = 17 [7%]	<i>k</i> = 32 [14%]
Reliability	IM = 0.36/0.31	IM = 0.32
Reported	<i>k</i> = 61 [27%]	<i>k</i> = 117 [52%]
Unreported	<i>k</i> = 45 [20%]	<i>k</i> = 98 [43%]
Sample size	IM = 0.24/0.36	IM = 0.18
<100	<i>k</i> = 45 [20%]	<i>k</i> = 80 [35%]
100–400	<i>k</i> = 55 [24%]	<i>k</i> = 119 [52%]
>400	<i>k</i> = 13 [6%]	<i>k</i> = 27 [12%]
Setting	IM = 0.11/0.17	IM = 0.49
Prison	<i>k</i> = 63 [28%]	<i>k</i> = 134 [59%]
Hospital	<i>k</i> = 42 [19%]	<i>k</i> = 71 [31%]
Unclear	<i>k</i> = 1 [0.4%]	<i>k</i> = 10 [4%]
Follow-up	IM = 0.07/0.15	IM = 0.28
<1 year	<i>k</i> = 11 [5%]	<i>k</i> = 21 [9%]
1–5 years	<i>k</i> = 43 [19%]	<i>k</i> = 83 [37%]
>5 years	<i>k</i> = 32 [14%]	<i>k</i> = 69 [30%]
Unclear	<i>k</i> = 24 [11%]	<i>k</i> = 54 [24%]
Age	IM = 0.17/0.14	IM = 0.11
≤35 years	<i>k</i> = 49 [22%]	<i>k</i> = 101 [44%]
>35 years	<i>k</i> = 37 [16%]	<i>k</i> = 70 [31%]
Unclear	<i>k</i> = 20 [9%]	<i>k</i> = 45 [20%]
Sex	IM = 0.82/0.44	IM = 0.73
Females	<i>k</i> = 14 [6%]	<i>k</i> = 20 [9%]
Males	<i>k</i> = 83 [38%]	<i>k</i> = 183 [84%]
Mixed	<i>k</i> = 14 [6%]	<i>k</i> = 18 [8%]
Males and Mixed	<i>k</i> = 97 [43%]	<i>k</i> = 201 [89%]

Note. Listed are the characteristics of the 12 study-, sample-, and individual-specific moderators evaluated in the updated meta-analysis concerning the criterion validity of the PCL scales' Factor 1 (F1), Factor 2 (F2), and the total score. The numbers and percentages of samples per moderator subgroup (*k* [%]) are reported as out of the total 217 samples. Listed are also the importance values (IM) for each moderator derived from the multimodel moderator analysis representing the overall support for a given moderator across all candidate models and all outcomes. Details on the multimodel analysis are mentioned in the main text and otherwise provided in the [online supplementary materials \(Section S4\)](#).

Country of origin was suggested as the strongest predictor moderating the criterion validity of the PCL scales. The corresponding importance values were high for the total score (IM = .99) and F2 (IM = 1.00). Both the total score and F2 showed a tendency for smaller validity in the United States compared to Canada and Europe when assessed across all outcomes. Statistical significance for this observation was found for institutional violence. In particular, the total score

for institutional violence was suggested to have smaller criterion validity in the United States ($d = 0.01$, 95% CI [−0.04, 0.05], $p = .772$) compared to Canada ($d = 0.36$, 95% CI [0.15, 0.58], $p < .001$; pairwise $z = -3.28$, $p = .006$) and Europe ($d = 0.42$, 95% CI [0.21, 0.64], $p < .001$; pairwise $z = -3.91$, $p < .001$). Likewise, F2 had smaller criterion validity for institutional violence in the United States ($d = 0.07$, 95% CI [−0.15, 0.28], $p = .551$) compared to Canada ($d =$

0.62, 95% CI [0.37, 0.89], $p < .001$; pairwise $z = -3.46$, $p = .003$) and Europe ($d = 0.81$, 95% CI [0.42, 1.31], $p < .001$; pairwise $z = -3.61$, $p = .002$).

Sex was suggested as the second strongest predictor moderating the criterion validity of the PCL scales. The corresponding importance value across all outcomes was high concerning F1 (IM = .82). There was a tendency for the criterion validity of F1 to be higher in female compared to male samples. Statistical significance for this observation was found for general and violent recidivism. In particular, F1 was suggested to have higher criterion validity in females compared to the category of *males and mixed* samples for general recidivism (females $d = 0.26$, 95% CI [0.02, 0.50], $p = .030$; males and mixed $d = 0.01$, 95% CI [-0.04, 0.06], $p = .605$; pairwise $z = -2.03$, $p = .042$) and violent recidivism (females $d = 0.24$, 95% CI [0.02, 0.46], $p = .030$; males and mixed $d = 0.01$, 95% CI [-0.02, 0.04], $p = .608$; pairwise $z = -2.08$, $p = .038$). The pairwise comparison between female and male samples (without the mixed samples) did not reach statistical significance.

None of the remaining moderators revealed significant subgroup contrasts, indicating that they did not moderate the PCL's criterion validity in the present multimodel analysis.

Publication bias

The results of the assessment of potential publication bias using Egger's test ($z = -2.21$, $p = .027$) and Begg and Mazumdar's test for rank correlation ($\tau = 0.2$, $p < .001$) assessing funnel plot asymmetry were both significant, supporting the concern of potential publication bias. The funnel plot showed a slight asymmetry in the upper parts of the plot, indicating that larger studies with more positive effects appeared to be slightly more likely to be published. The p-curve analysis, however, showed no indication of publication bias, as expressed by a negatively sloped (Flatness test $p > .9999$) and right-skewed (Right-skewness test $p < .0001$) p-curve. As a right-skewed p-curve has been suggested to potentially indicate p-hacking in observational compared to experimental research (Bruns & Ioannidis, 2016, however, publication bias may still threaten the validity of the updated meta-analyses. Both the funnel plot and the p-curve are provided in the [online supplementary materials](#) (Section S5).

Discussion

Psychopathy is a clinical construct with a long but checkered history. The construct has been both hailed as a "unified theory of crime" (DeLisi, 2009, p. 256) and bemoaned as an

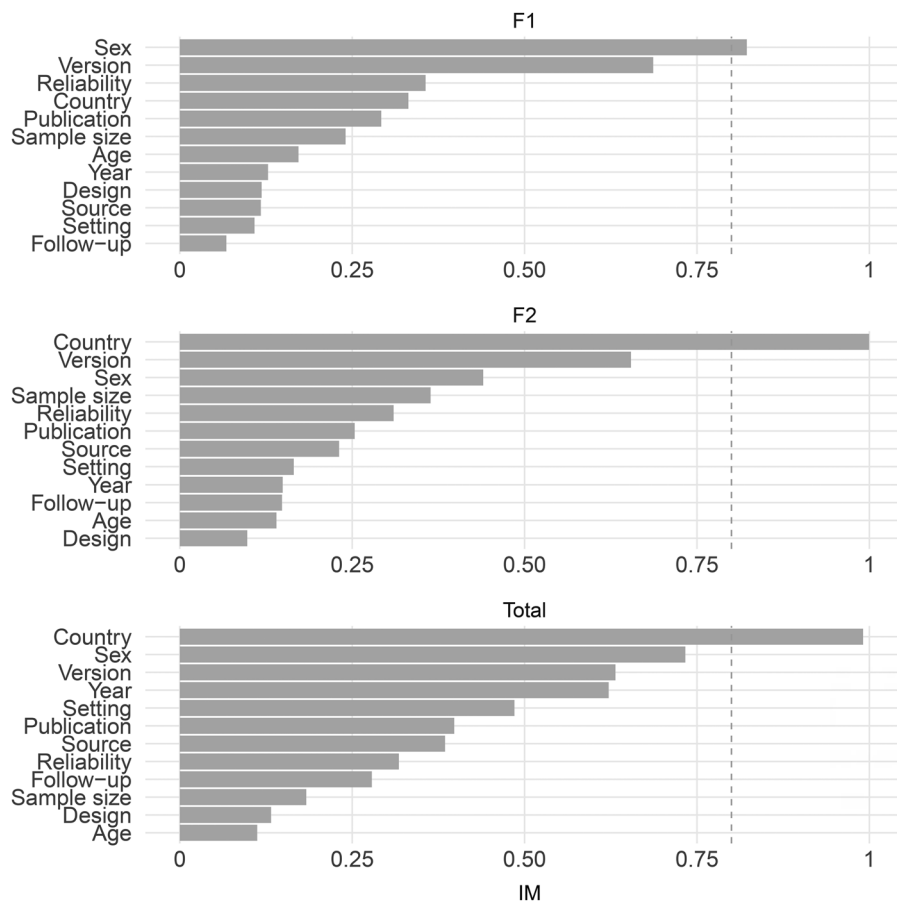


Figure 2. Moderator Importance. Note. Plots illustrating the model-averaged importance of each moderator concerning the criterion validity of PCL scales' Factor 1 (F1), Factor 2 (F2), and the total score obtained in the updated meta-analysis. Importance values (IM) represent the overall support of each moderator, i.e., the sum of weights across all models in which a moderator appeared. A vertical dashed line is drawn at IM = .8, sometimes used as a cutoff to differentiate between important and less important variables, though this is an arbitrary division (Anderson, 2007; Calcagno & Mazancourt, 2010). Details on the multimodel analysis are mentioned in the main text and otherwise provided in the [online supplementary materials](#) (Section S4).

“elusive concept” (Cooke, 2018, p. 15). Undoubtedly, the PCL-R and its derivatives have become frequently used instruments for risk assessment within the criminal justice system (Hare, 1996; Hemphill & Hare, 2004). Using the PCL instruments within forensic settings may inform sentencing and release decisions or security-level prison classifications. Its misuse, however, may influence the criminal justice system, even regarding the imposition of the death penalty (DeMatteo et al., 2020b; Hare et al., 2020). Given the impact of the construct of psychopathy on real-world decision-making, research in the field requires continuous updates. The present meta-analysis provides an update on the current evidence regarding the criterion validity of the PCL instruments as particularly frequently used measures of said construct within legal contexts. Although this update is close in time to the recent meta-meta-analysis by DeMatteo and Olver (2022), a considerable number of additional eligible studies ($k=46$ samples, 11,585 individuals) could be included that were not covered by DeMatteo and Olver (2022). Including these additional samples, the updated meta-analysis represents, to our knowledge, the largest meta-analysis of the criterion validity of the PCL considering both recidivism and institutional outcomes ($k=217$ samples, 46,857 individuals). Together with the additional moderator analysis, the present analysis contributes essential findings regarding the cross-outcome, cross-cultural, and cross-gender generalizability of the criterion validity of the PCL scales.

Criterion validity

The updated meta-analysis yielded evidence for considerable variation in the criterion validity of the PCL scales concerning its factors and targeted outcomes. Considering the factor level, the PCL total score and F2 had moderate, and F1 had small criterion validity across all outcomes. Although these findings were generally consistent with the meta-meta-analysis by DeMatteo and Olver (2022), the updated meta-analysis provided evidence for statistically significant differences at the factor level. F2 was statistically more predictive than the total score and F1, and F1 was statistically less predictive than the total score and F2. These findings suggest that the criterion validity of the PCL scales is almost entirely attributable to the social deviance reflected in F2. In contrast, the core psychopathic personality features reflected in F1 reduce it. This observation has been recognized in previous meta-analyses, such as by Yang et al. (2010), which pointed to F2 as a powerful predictor of recidivism. In contrast, the link between F1 and future criminality was less pronounced. Similarly, Camp et al. (2013) called into question the controversial description of psychopathic individuals as so-called “predators” (p. 467), given that the association with violence was more robust for antisocial and impulsive tendencies. Consequently, one has to distinguish between the original purpose of the PCL instruments as diagnostic measures of a personality trait (i.e., its construct validity) and its use as risk-assessment tools (i.e., its criterion validity). While the complete PCL instruments appear more appropriate from the perspective of construct coverage, the criterion validity

in terms of predicting recidivism may be facilitated in the sense that less is more: The subtotal on F2 outperformed the total score (e.g., Harpur et al., 1989; Olderbak et al., 2021). Or, in the words of Cleckley (1941), criminal proclivity has been a hallmark of the psychopathy construct (Hare & Neumann, 2008), yet in risk assessment, this disposition appears to eclipse the deceitful and affectionless components of psychopathy. This is eventually in line with Meehl’s (1989) self-proclaimed malignant maxim: “The best way to predict anybody’s behavior is his behavior in the past.” (p. 532). The differential validity of the PCL-R total and factor sub-totals attests to the usefulness of treating the factor sub-totals separately for practical purposes in risk assessment. Moreover, the gradation of magnitude as described for the PCL-R total score (Hare, 2003) could easily be transferred to the factor sub-totals as well.

The factor level of the PCL scales may also be relevant when comparing their criterion validity to other risk instruments. For example, the comparative meta-analysis by Yang et al. (2010, p. 753) found the PCL-R to predict violence significantly worse than the Historical Clinical and Risk Management Scheme (HCR-20 V2; Webster et al., 1997), the latter which notably incorporated the PCL-R total score as one of its items. On the other hand, the Violence Risk Appraisal Guide (VRAG/VRAG-R; Harris et al., 1993; Rice et al., 2013), which initially also included the PCL-R total score in one of its items but later limited its contribution to half of F2 (i.e., the antisocial facet) upon its revision—a decision that only seemed straightforward considering the present results –, was shown to perform similarly to the PCL-R in violence prediction in the comparative analysis (Yang et al., 2010). Overall, however, the differences in predicting violence between the instruments mentioned above and others have been described as small and inconsistent and unlikely to translate into meaningful differences in practice (e.g., Campbell et al., 2009; Fazel et al., 2012, 2022; Ogonah et al., 2023; Singh et al., 2011; Walters, 2003a, 2006).

Considering the outcome level, the updated meta-analysis also suggested variations in the criterion validity of the PCL instruments between the type of recidivism and institutional outcomes. The PCL scales were found to perform reasonably well in predicting general recidivism, violent recidivism, and institutional outcomes, yet fared less favorably concerning sexual recidivism. Although most of the corresponding effect sizes observed in the updated meta-analysis for general and violent recidivism turned out to be moderate, those for the institutional outcomes fell slightly short of being moderate, and the predictive value for sexual offenses was poor. These findings strengthen the recommendation that the PCL instruments can be used confidently for predicting general and violent recidivism, in line with the meta-meta-analysis by DeMatteo and Olver (2022) as well as other previous research (e.g., Larsen et al., 2020; Olver & Riemer, 2021).

Regarding institutional outcomes, however, the present findings suggest that the PCL instruments should be used cautiously for prediction purposes, as has been debated elsewhere, including our research (DeMatteo et al., 2020a, 2020b; Olver et al., 2020). Particularly, the effect size of the criterion validity of the PCL-R for institutional violence observed

in this meta-analysis ($d=0.49$) should be interpreted concerning the threshold of risk assessment tools for the criterion validity of at least $d=0.358$ set by DeMatteo and Olver (2022). While DeMatteo et al. (2020) argued for a moratorium on the PCL-R in contexts of institutional violence because of low criterion validity, our results confirm caution in using the PCL-R for this purpose.

Concerning the risk of sexual (and IPV) recidivism, the PCL instruments ought not to be used as the sole predictor. Still, they may preferably be used in combination with other static or dynamic risk assessment tools focusing specifically on sexual reoffenses, such as the Static-99R (Phenix et al., 2016), if found to be valid in combination with the STABLE-2007 (Hanson et al., 2007), or the Sex Offender Risk Appraisal Guide (SORAG; Quinsey et al., 2006). Together, the observations made in the updated meta-analysis corroborated the strengths and shortcomings of the PCL-R in predicting adverse outcomes of different kinds.

Moderator analysis

The updated meta-analysis also provided a moderator analysis. Although the results of the moderator analysis were consistent with past findings overall, they highlight potential limitations in the PCL's cross-cultural and cross-gender generalizability. Due to the traditional meta-analytical approach compared to the meta-meta-analysis by DeMatteo and Olver (2022) and the increase in sample size compared to other previous traditional meta-analyses, the updated meta-analysis provides greater precision and robustness regarding the current evidence on the predictors moderating the criterion validity of the PCL scales.

One robust effect that explained some of the heterogeneity about psychopathy was the country in which the PCL assessment was conducted. The most robust observation was the smaller predictive effect observed in studies conducted in the United States compared to Canada and Europe. Although earlier research on the PCL mainly conducted in North America showed evidence of good cross-cultural generalizability (e.g., Cooke, 1998; Gonçalves, 1999; Grann et al., 1998; Moltó et al., 2000), more recent research was more consistent with the present findings suggesting smaller effects in the United States (Guy et al., 2005; Leistico et al., 2008). This may indicate that PCL measures in institutions of the United States may have been less accurate in identifying individuals particularly prone to reoffend than those outside the United States (Guy et al., 2005). The weaker effects in the United States presumably result from methodological differences, such as the quality of PCL ratings, the definition of recidivism, follow-up length, base rates, or racial and ethnic compositions of the samples (e.g., Grann et al., 1999; Leistico et al., 2008). Cultural bias in PCL ratings could also stem from differences in the recognition of the latent traits of psychopathy. Some traits, such as the deficient affective experience, have been suggested to have greater cross-cultural stability compared to others and, therefore, have been considered the pan-cultural core of psychopathy (Cooke et al., 2005). Authorship bias, as described for other risk instruments (Singh et al., 2013), may also be

suggested as most researchers involved in developing the PCL scales were based in Canada and strongly influenced Canadian forensic research. The updated meta-analysis did not assess authorship bias and could not provide further support for such a bias.

Another robust effect, consistent with past findings on the PCL scales, was found concerning the biological sex of persons with offense histories. Previous work already called into question the cross-gender applicability of the PCL instruments. Despite good support for the reliability of the PCL-R in women, support for its criterion validity compared to men has been equivocal (e.g., Dolan & Völlm, 2009). Though psychopathy has been suggested as an essential risk factor for re-offending in women, the effect may not be as strong as it is for men (e.g., Nicholls et al., 2005). Some explanations for these gender differences were related to a generally lower prevalence of psychopathy and lower relapse rates (Grann, 2000; Jackson et al., 2002; Salekin et al., 1997; Warren et al., 2003). Another explanation could be that psychopathy is expressed differently in women than men, for example, in terms of manipulative behavior, violation of social norms, or various forms of aggression (e.g., Neumann et al., 2007; Verona & Vitale, 2006; Wynn et al., 2012). It has, therefore, been suggested that certain PCL-R items might not be adequately assessing the psychopathy construct in women (Forouzan & Cooke, 2005; Weizmann-Henelius et al., 2010). For example, concerning the factor level comparing the core psychopathic personality traits (F1) and the social deviance component (F2), some studies suggested F1 as a more robust predictor and more informative in women compared to F2 (Leistico et al., 2008; Yang et al., 2010). In contrast, the opposite was suggested for men (Cale & Lilienfeld, 2002; Neumann et al., 2007; Neumann & Hare, 2008; Richards et al., 2003; Salekin et al., 1998). Others noted that F2 was more closely associated with violence in women than F1 and that both were equally related to criminal convictions, in contrast to common findings for males (Loucks & Zamble, 2000). Some studies reported no sex-related effects (Guy et al., 2005; Walters, 2003b). The updated meta-analysis supported the first view, with F1 being associated with higher criterion validity in females than males (at least concerning our *males and mixed* category) for general and violent recidivism. In contrast, no such effects were observed for F2.

Another issue of generalizability concerns the debate on potential differences in the PCL measures depending on the composition of the racial sample. Previous research reported inconclusive evidence. While some studies reported differences in the mean PCL:YV score between black and white youths (McCoy & Edens, 2006), others did not find differences in the mean PCL-R score between black and white adults (Skeem et al., 2004) or Caucasian and African American adults (Cooke et al., 2001). While some studies supported the existence of racial differences in the PCL-R's criterion validity (Baskin-Sommers et al., 2013; Lynn, 2002), others criticized these findings (Skeem et al., 2003) or encountered difficulties in separating the influences of race and country due to the presence of multicollinearity between these two moderators (Leistico et al., 2008). In the updated

dataset, race information was rarely reported in the primary studies, resulting in a large subgroup with unclear racial composition. As this would have made potential racial effects hard to interpret, race was not considered a moderator in the present analysis.

None of the remaining moderators (publication status, publication year, design, source, version, reliability, sample size, setting, follow-up, age) were found to be predictive concerning the PCL scales in the present analysis. The present work, therefore, did not support earlier findings that the PCL-R total score and F2 had higher criterion validity when ratings were based on file review only compared to a combination of file and interview information (Leistico et al., 2008), that F2 had more considerable criterion validity in forensic-psychiatric samples compared to prisoners (Leistico et al., 2008), or that F2 had more considerable criterion validity with increasing follow-up length (Leistico et al., 2008). Other predictors, such as the publication status (published vs. unpublished) (Guy et al., 2005), the study design (prospective vs. retrospective), the PCL version (PCL-R vs. PCL:SV) (Leistico et al., 2008), or the age (adults vs. juveniles) (Guy et al., 2005; Leistico et al., 2008; Walters, 2003b) were also previously not reported to significantly affect the criterion validity of the PCL scales.

Limitations

The updated meta-analysis has other shortcomings besides the limitations already mentioned. Even though we strived to avoid sample overlap when collecting the primary studies in the updated dataset, some overlap certainly occurred. Sample overlap occurs when multiple estimates are drawn from the same set of individuals, or a subset thereof, in different primary studies. To overcome this limitation, the cluster-robust inference method was implemented. This aspect should be considered when interpreting the present findings.

The present meta-analysis was limited to evaluating the main moderator effects but did not assess interaction effects between moderators. Concerning the two moderators country and sex, for instance, an interaction analysis might have provided more insight. However, because of the relatively small number of female samples in the updated dataset, most of the corresponding interaction subgroups would have been extremely small, making the interpretation of interaction effects uncertain. In addition, assessing interaction effects within a multimodel framework increases the number of candidate models exponentially, thus rendering the interpretation of numerous interaction terms complex.

The updated meta-analysis did not address psychiatric comorbidities. Psychiatric conditions were rarely coded in the included studies, and if so, primarily within forensic-psychiatric settings ($k=33\%$ of the studies). Evaluating such a relatively small sample size with potentially multiple subgroups of psychiatric diagnoses would have been inappropriate in the context of the updated meta-analysis. Future meta-analyses should address whether and how psychiatric diagnoses may influence the criterion validity of the PCL scales.

The present work did not provide a risk of bias assessment for the included studies. This may limit the conclusions to be drawn regarding the applicability of the updated meta-analysis's findings.

Practical implications

The PCL instruments are the primary assessment tools for measuring psychopathy in correctional and legal contexts. Considering the findings of the updated meta-analysis, the following practical implications should be considered.

First, although one should acknowledge that the PCL scales were developed as a psychodiagnostic measure, not as a risk assessment tool, the present findings on its criterion validity may guide its use in future risk assessment. Therefore, we briefly illustrate an example for interpreting its effect size estimate in terms of Cohen's d , which essentially expresses the mean difference between two groups: recidivists and non-recidivists. Let us use the mean and SD estimates provided for North American male offenders ($N=5,408$) in the PCL-R manual as a frame of reference (Hare, 2003). For simplicity's sake, let us also assume equally sized groups of recidivists and non-recidivists and normal distributions for both. Then, the d for the total score across all outcomes (i.e., $d = 0.55$; cf. Table 1) would imply that 70.9% of the "recidivists" will be above the mean of the "non-recidivists," 78.3% of the two groups will overlap. There is a 65.1% chance that an individual picked at random from the "recidivists" group will have a higher score than an individual picked at random from the "non-recidivists" group (for details, see Grice & Barrett, 2014; Magnusson, 2023).

Furthermore, the inter-rater reliability in PCL-R ratings needs to be addressed. Over more than 15 years, dating back to the first study on compromised inter-rater reliability in PCL-R ratings by Murrie et al. (2008), a growing body of literature addressed this issue, especially in adversarial legal systems. Although some of the inter-rater discrepancies observed may be due to biased raters and expectancy effects (e.g., Murrie et al., 2008; Murrie et al., 2013), the fact remains that in many cases, there are pronounced differences in the PCL-R scores provided by different raters for the same individual, at least within the field (rather than research) context (DeMatteo & Olver, 2022). Therefore, we should point out as a caveat that the effect sizes reported herein likely represent upper-limit estimates of the magnitude that can be expected in forensic-psychiatric reports, legal proceedings, or correctional decisions. In other words, the real-world effect sizes are likely smaller than the magnitude reported in the present analysis. DeMatteo and Olver (2022) provided recommendations to address the problem of imperfect inter-rater reliability of the PCL-R by using percentile ranks and SDs of the mean when reporting individual PCL-R scores, which both account for the dimensionality of the psychopathy construct and may thus augment interpretation.

Last, as previously mentioned by DeMatteo and Olver (2022), though the present analysis focused on the predictive properties of standalone PCL scales, they are preferably used in combination with other formal risk assessment measures

specific to the outcome being evaluated, such as the HCR-20 (Douglas et al., 2013), the VRAG-R (Rice et al., 2013), the Sex Offender Risk Appraisal Guide (SORAG; Quinsey et al., 2006), the Violence Risk Scale (VRS; Wong & Gordon, 2006), or the Sexual Offender version of the VRS (VRS-SO; Olver et al., 2007). In this context, the incremental validity of the PCL instruments in combination with other instruments has also been examined in several studies (e.g., Yoon et al., 2022).

Conclusions

This updated meta-analysis represents the most extensive synthesis of the current evidence on the criterion validity of the PCL scales regarding recidivism and institutional outcomes in legal contexts. The results support the meta-meta-analysis by DeMatteo and Olver (2022) but provide greater robustness of the predictive effects due to a traditional meta-analytical approach, a considerably larger number of samples included, and the contribution of a moderator analysis. The heterogeneous nature and the limitations of the present findings still complicate concluding the cross-outcome, cross-cultural, and cross-gender generalizability of the criterion validity of the PCL scales. However, the present findings may support researchers and decision-makers in refining the PCL instruments as risk assessment tools in the criminal justice system.

Disclosure statement

The authors A. Mokros and E. Habermeyer receive royalties from the German version of the Hare Psychopathy Checklist-Revised (PCL-R) manual. Otherwise, they declare no conflicts of interest.

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Data availability statement

The dataset generated for the research has been made available in The Open Science Framework (OSF) (Holper, L. and Ehrbar, C. and Fancellu, F. and Hauser, Nicole C. and Habermeyer, E. and Mokros, A., 2023; <https://osf.io/y6dtx/>).

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